

(LRC) was defined as complete imaging response or cessation of vaginal bleeding if MRI was unavailable.

Results: From 2007–2016 45 pts underwent definitive BT alone to a median dose of 37.5 Gy (range, 35–45 Gy) in 5–6 fractions. Nineteen pts (42%) were treated with MRI-based plans with MRI obtained with applicator in place, 19 (42%) with CT-based plans with MRI obtained prior to BT, and 7 (16%) with CT-based plans alone. The median GTV and HR-CTV volumes were 5.9 cc (range, 0.7–18.7 cc) and 80.9 cc (range, 17.2–159.0 cc), respectively. The median cumulative dose to 90% (D90) of the GTV was 132.8 Gy (range, 76.5–295.6 Gy) EQD2, and the median HR-CTV D90 was 49.7 Gy (range, 34.5–57.2 Gy). The median follow-up among living pts was 18.6 months (range, 3.0–64.3 months). Cessation of vaginal bleeding occurred in 98%. Post-treatment MRI was performed in 29 pts (64.4%), of which complete response (CR) was demonstrated in 26 (90%), and 3 (10%) had residual nonspecific signal abnormality. Of these 3 pts, 2 had resolution of vaginal bleeding and no further evidence of disease on follow-up, and the 3rd had residual endometrial thickening and a suspicious iliac node on follow-up MRI and subsequently developed peritoneal carcinomatosis 9 months after BT. Among pts with imaging CR, there were 2 local failures consisting of nodularity within the endometrium managed with D&C and levonorgestrel IUD in 1 pt and a vaginal recurrence salvaged with vaginal BT in another. The GTV D90 were 104.3 Gy and 126.1 Gy in these 2 pts, compared with a median of 138.4 Gy in other patients. The 2-year LRC and overall survival rates were 90% and 86%, respectively. Among a subset of 20 pts with at least 2 years follow-up (median 37.6 months), no additional local failures were seen beyond 2 years. The median rectum, sigmoid, and bladder dose to 2 cc (D2cc) EQD2 values were 19.8 Gy (range, 8.3–46.3 Gy), 40.4 Gy (range, 17.5–60.7 Gy), and 45.8 Gy (22.1–66.1 Gy), respectively. No grade 3 or greater acute or late toxicity was observed.

Conclusions: Image-guided HDR BT alone for treatment of early-stage, medically inoperable endometrial cancer is feasible with excellent response rates and early clinical results. While dose to the HR-CTV was relatively low, the high LRC rates may be related to the high dose delivered to the GTV with 3D planning. This image-guided approach also allows better sparing of adjacent critical organs and ensures coverage of the target, which likely contributed to the low toxicity rate and high rate of local control in comparison with prior series utilizing 2D point-based planning. These data suggest that in select pts with grade 1–2, low-volume disease, EBRT may be omitted without compromising disease control and thus sparing the potential morbidity of pelvic radiation in a medically frail population.

PLENARY SESSION

Friday, April 21, 2017

1:45 PM - 3:00 PM

PL01 Presentation Time: 1:45 PM

Long Term Outcomes of I¹²⁵ Eye Plaque

Brachytherapy in Patients with Choroidal Melanoma

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Purpose: To determine the relationship between MCW prescription parameters and outcomes in patients with choroidal melanoma treated with I¹²⁵ eye plaque radiotherapy. Background: Choroidal melanoma is a rare condition that impacts vision and survival. Eye plaque therapy is a vision-preserving treatment that has been utilized for many decades; however, further optimization of the treatment parameters is needed to reduce toxicities and improve outcomes. Variations in treatment approaches exist among different institutions. At MCW the prescription dose is typically specified at the tumor apex and the prescription dose and dose rate are slightly lower than used in COMS studies. This retrospective review investigated the relationship between MCW treatment parameters and outcomes.

Materials and Methods: From 2000–2015, 134 consecutive patients underwent I¹²⁵ eye plaque (COMS design) brachytherapy for treatment of choroidal melanoma. 3 patients were excluded due to follow up <1 year. With IRB approval, data was collected via retrospective review, including

demographic, clinical and dosimetric parameters (total dose, dose rate, dose gradient across the lesion). Dose and dose rate to macula and optic disk were calculated accounting for uncertainty of the rotational position of the plaque. Simple statistical analysis was performed to assess outcomes including decrease in visual acuity by >2 lines on a Snellen chart, retinopathy, functional vision preservation at 2 and 5 years, local and distal disease control, and survival (i.e., disease-free survival and overall survival).

Results: Population consisted of 98% Caucasian with median age of 62 (26–88). 53% were women. Median tumor height was 3.4 mm (1.5–10.47) with median basal diameter of 10.7 mm (3.2–18.6). COMS stage was 18% small and 76% medium. Average plaque size was 16 (12–20). Median dose(TG43) received at the apex was 68.7 (51.8–77.7) Gy and dose rate 58.35 (46.8–70.4) cGy/hr. Median dose received at the base was 129 (79.8–350) Gy and dose rate 110.7 (61.1–299) cGy/hr with the gradient across the tumor 62.3 (11.9–280) Gy. With median follow up of 47 months local recurrence was observed in 13% and metastatic disease in 17% of patients. Median dose and dose rate received at the tumor apex in patients with local recurrence was not statistically different than in patients without local recurrence, 68.8 Gy (58.6 cGy/hr) vs. 68.7 Gy (58.3 cGy/hr) respectively. Metastases to liver was the most common site with median time to diagnosis from brachytherapy procedure of 22 months. Median distance to fovea was 6 mm and to optic nerve was 6.5 mm. Median dose received by fovea was 33.95±13 (3.55–307)Gy and dose rate 28.6±10.2 (3.15–259) cGy/hr. Median dose received by optic disk was 29.5±10.8 (3.55–307)Gy and dose rate 24.6±9 (3.15–251) cGy/hr. 27% of patients had 20/20 vision and 12% had significant visual impairment pretreatment. 105 and 48 patients had vision assessed post treatment at 2 and 5 years respectively with >2 lines drop on Snellen chart seen in 43% at 2 years and 46% at 5 years. Severe visual loss (>20/200) was seen in 33%. Vision loss potentially induced primarily by radiation was seen in 29%. 2 patients have undergone enucleation due to eye pain and 9% for local recurrence.

Conclusions: Patients with choroidal melanoma treated with I¹²⁵ eye plaque brachytherapy at our institution experienced comparable local control and rate of metastases, and had better visual preservation in comparison to reported by COMS studies. Dose specification at the apex does not result in an increased risk of recurrence in patients with apical heights <5mm. The total dose and dose rate used may improve long term visual outcome as compared to the COMS study.

PL02 Presentation Time: 2:00 PM

Evaluation of a Machine-Learning Algorithm for Treatment Planning in Prostate Low-Dose-Rate Brachytherapy



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Purpose: This retrospective study presents the results of a machine-learning (ML) algorithm used to automatically generate high-quality, prostate low-dose-rate (LDR) brachytherapy treatment plans. The algorithm can replicate several characteristics of expert created pre-operative LDR treatment plans. The planning efficiency, dosimetry, and quality (as assessed by experts) of pre-operative plans generated with an ML model was evaluated.

Materials and Methods: A training database of 100 prostate LDR treatment plans - filtered for high-quality post-operative dosimetry - was used to train an ML model to predict source patterns that approximate successful plan dosimetry. The algorithm was trained on cases with similar anatomic features to obtain a starting treatment plan. A stochastic, iterative decision process was then used to examine different plan states to find an optimal state for the current anatomy; planning strategies were